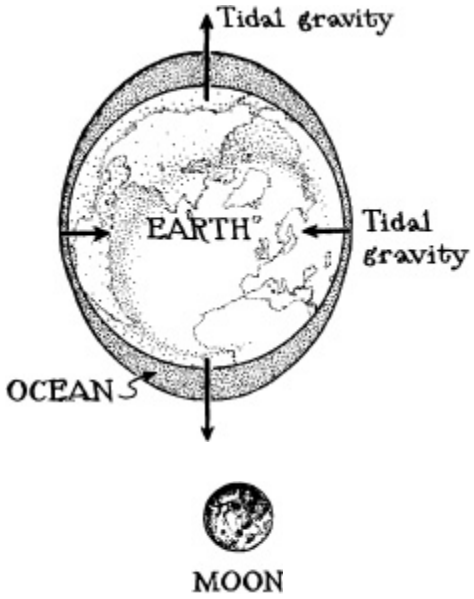


On the side of the Earth nearest the Moon, the lunar gravity is stronger than at the Earth's center, so it pulls the oceans toward the Moon more strongly than it pulls the solid Earth, and the oceans in response stretch outward a bit toward the Moon. On the side farthest from the Moon, the lunar gravity is weaker, so it pulls the oceans toward the Moon less strongly than it pulls the solid Earth, and the oceans in response stretch out away from the Moon. On the left side of the Earth, the Moon's gravitational pull, which points toward the Moon's center, has a slight rightward component, and on the right side it has a slight leftward component; and these components squeeze the oceans inward. This pattern of oceanic stretch and squeeze produces two high tides and two low tides each day, as the Earth rotates

If the tides at your favorite ocean beach do not behave in precisely this way, it is not the fault of the Moon's gravity; rather, it is because of two effects: (1) There is a lag in the water's response to the tidal gravity. It takes time for the water to move in and out of bays, harbors, river channels, fjords, and other indentations in the coastline. (2) The Sun's gravitational stretch and squeeze are about half as strong on the Earth as the Moon's, but are oriented differently because the Sun's position in the sky is (usually) different from the Moon's. The Earth's tides are a result of the combined tidal gravity of the Sun and the Moon.